¿Cómo influye la filosofía estadística en nuestro trabajo?

● Tu filosofía estadística (consciente o inconscientemente) influye en gran medida en todas las partes de tu flujo de trabajo científico:

○ Los tipos de teorías y preguntas que pones a prueba (si solo conoces test de significancia estadística basada en hipótesis nulas,
solo plantearás preguntas que puedan responderse mediante test de significancia estadística basada en hipótesis nulas)

○ Diseño experimental

○ Análisis de datos y verificación de modelos

○ Interpretación de los resultados (tanto los tuyos como los de literatura)

○ Comunicación científica (visualización de datos y modelos, presentación de los resultados)

● La inferencia bayesiana te ofrece una forma fundamentada de abordar todos los componentes mencionados anteriormente

Dos interpretaciones de la probabilidad

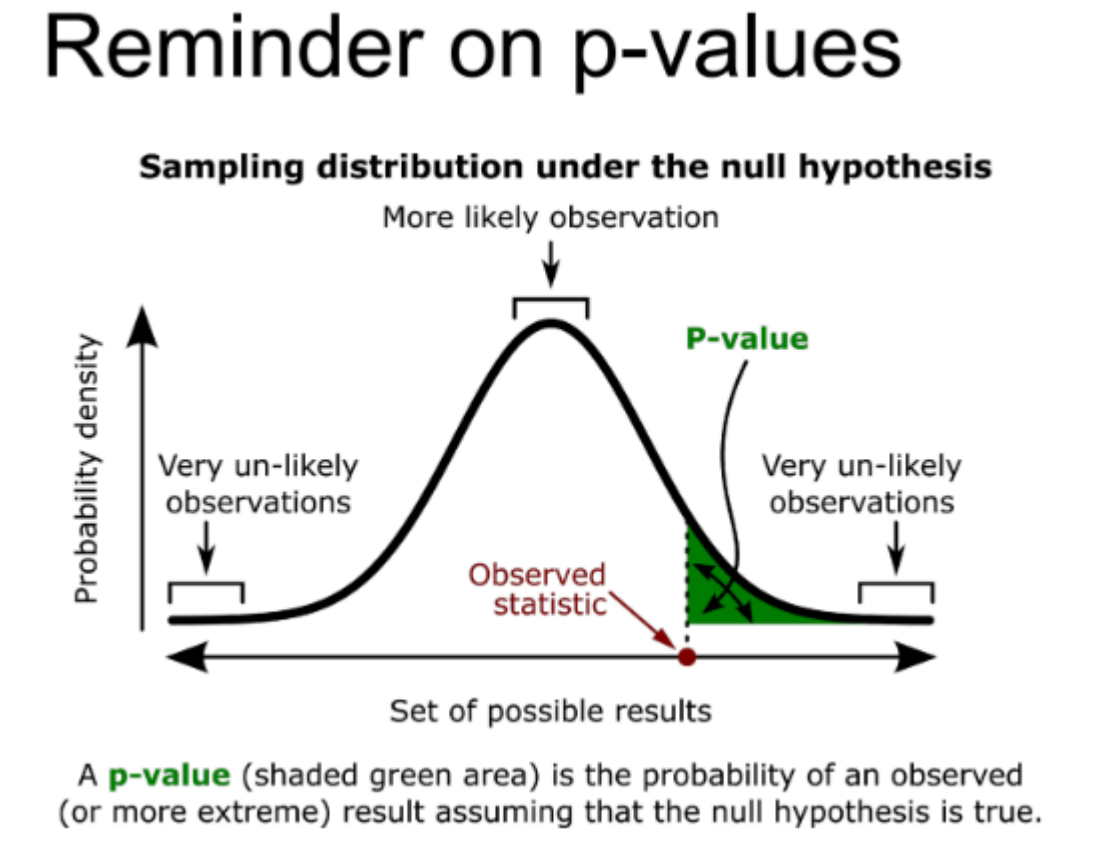
● Teoría de la probabilidad: reglas matemáticas para manipular probabilidades
● La teoría de la probabilidad no genera controversias, pero sí lo hace su aplicación al mundo real; dos interpretaciones de la probabilidad:
○ Probabilidad aleatoria/frecuentista: frecuencia esperada tras numerosas repeticiones de un procedimiento

○ Probabilidad bayesiana: grado de creencia que se debe asignar a un evento o proposición (que depende intrínsecamente del estado de conocimiento del agente)

● ¿Por qué es importante esto?

Llegas a tu casa y ves en el portal el suelo mojado.

¿Cuál es la probabilidad de que haya llovido?



Testing our intuitions about p-values

Suppose you have a treatment that you suspect may alter performance on a certain task. You compare the means of your control and experimental groups (say 20 subjects in each sample). Further, suppose you use a simple independent means t-test and your result is (t = 2.7, d.f. = 38, p = 0.01). Please mark each of the statements below as "true" or "false". "False" means that the statement does not follow logically from the above premises. Also note that several or none of the statements may be correct.

1) You have absolutely disproved the null hypothesis (that is, there is no difference between the population means). [ ] true / false [ ]

2) You have found the probability of the null hypothesis being true. [ ] true / false [ ]

3) You have absolutely proved your experimental hypothesis (that there is a difference between the population means). [ ] true / false [ ]

4) You can deduce the probability of the experimental hypothesis being true. [ ] true / false [ ]

5) You know, if you decide to reject the null hypothesis, the probability that you are making the wrong decision. [ ] true / false [ ]

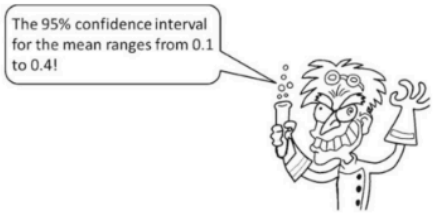
6) You have a reliable experimental finding in the sense that if, hypothetically, the experiment were repeated a great number of times, you would obtain a significant result on 99% of occasions. [ ] true / false [ ]

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Y ahora, intervalos de confianza.

Testing our intuitions about confidence intervals

Professor Bumbledorf conducts an experiment, analyzes the data, and reports:



Please mark each of the statements below as "true" or "false". False means that the statement does not follow logically from Bumbledorf's result. Also note that all, several, or none of the statements may be correct:

1. The probability that the true mean is greater than 0 is at least 95%. ☐True ☐False
2. The probability that the true mean equals 0 is smaller than 5%. ☐True ☐False
3. The "null hypothesis" that the true mean equals 0 is likely to be incorrect. ☐True ☐False
4. There is a 95% probability that the true mean lies between 0.1 and 0.4. ☐True ☐False
5. We can be 95% confident that the true mean lies between 0.1 and 0.4. ☐True ☐False
6. If we were to repeat the experiment over and over, then 95% of the time the true mean falls between 0.1 and 0.4. ☐True ☐False

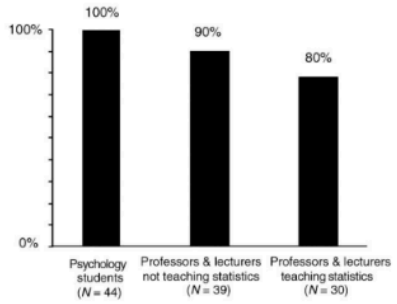
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¿En cuántas preguntas has seleccionado una respuesta correcta?

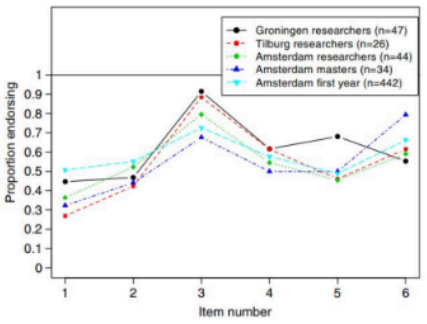
Pues…, ¡vaya!

Testing our intuitions about frequentist results

- If you endorsed any of the previous statements, **the good news** is that you are in good company!



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- **The bad news** is that none of the statements are correct!

Tranquilo, muchos profesores de estadística han fallado las preguntas 🙂